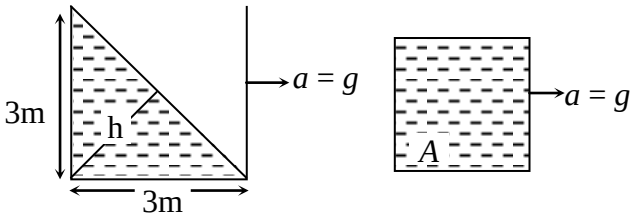
PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $P_1 = \rho\sqrt{g^2 + a^2}\,h = \rho\sqrt{2g}\,\frac{3}{\sqrt{2}} = 3\rho g$

$P_2 = \rho g h' + \rho a l = 3\rho g + 3\rho g = 6\rho g$

$\frac{P_1}{P_2} = \frac{1}{2}$

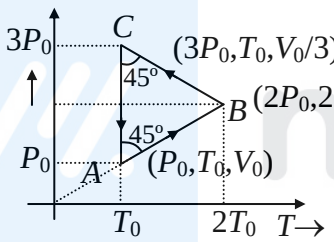


$\therefore$ **(d)**

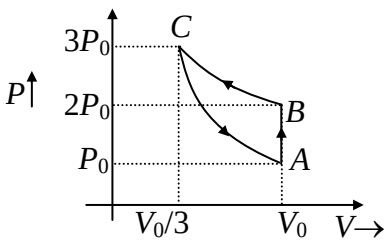
2. $1200 \times (g + a) = \frac{1}{3} \times 2.4 \times 10^8 \times 3 \times 10^{-4}$ or $a = 10\,\text{ms}^{-2}$

$\therefore$ **(b)**

3. AB is an isochoric process and CA is an isothermal process



Corresponding P-V curve for the above cyclic process.



$\therefore$ **(b)**

4. According to the principle of conservation of angular momentum

$m\sqrt{\frac{10GM}{6R}}R = mv'(R + h)$ … (1)

Applying principle of conservation of mechanical energy

$-\frac{GMm}{R} + \frac{1}{2}m\left(\frac{10GM}{6R}\right) = -\frac{GMm}{R+h} + \frac{1}{2}mv'^2$ … (2)

Solving the above equations we get $h = 4R$

$\therefore$ **(c)**

5. The maximum and minimum frequency is received by the receiver when it is at the mean position.

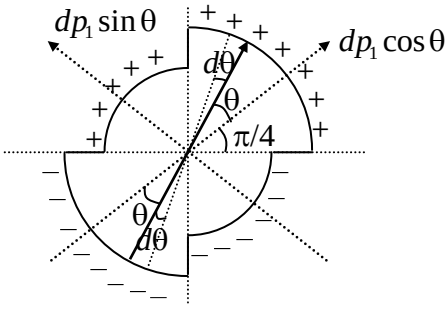
$$f_{max} = \left(\frac{v + v_0}{v}\right) f_0 = 1020\,\text{Hz}$$

$$f_{min} = \left(\frac{v - v_0}{v}\right) f_0 = 960\ \text{Hz}$$

Frequency band width received by the receives ranges between 960 Hz – 1020 Hz.

$\therefore$ **(c)**

6. Dipole moment of the two bigger arcs is



$$\int dP_1 = \int_{-\pi/4}^{\pi/4} dq(2\sqrt{3}R)\cos\theta$$

$$\int dP_1 = \int_{-\pi/4}^{\pi/4} 6\lambda R^2 \cos d\theta$$

$$P_1 = 6\lambda R^2 \int_{-\pi/4}^{\pi/4} \cos\theta d\theta$$

$$= 6\sqrt{2}\lambda R^2$$

Similarly for smaller arcs the dipole moment is

$$P_2 = 2\sqrt{2}\ \lambda R^2$$

$$|\vec{P}_{net}| = \sqrt{P_1^2 + P_2^2} = 4\sqrt{5}\lambda R^2$$

$\therefore$ **(a)**

7. Time during which the charged particle remains in magnetic field is

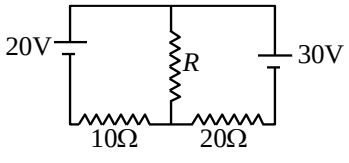
$$T = \frac{8\pi m}{3qB} = 4.15s$$

$$\delta = \frac{4\pi}{3} \Rightarrow \text{deviation of the charged particle}$$

Shortest distance between the entry and exit point is $= \sqrt{3}R = 6\sqrt{3}$ m

$\therefore$ **(b)**

8. The given figure can be redrawn is



For P_{max}, $R = \frac{10 \times 20}{10 + 20}\Omega = \frac{20}{3}\Omega$

$\therefore$ **(d)**

SECTION – II
Multiple Correct Choice Type

9. According to Newton's second law of motion

$$\tau = \frac{dL}{dt} \text{ or } \int_0^t \tau dt = L_f - L_i$$

$$\Rightarrow \quad \int 4qE\frac{a}{\sqrt{2}}dt = L_f$$

$$qa^2B_0 = 2ma^2\omega$$

$$\omega = \frac{qB_0}{2m}$$

$\therefore$ **(a, c, d)**

10. Maximum current through the inductor is $I_{\max} = \frac{Q_0}{\sqrt{LC}}$

When only switch k_1 is closed then $I_{\max} = \frac{10\times10^{-3}}{5\times2\times10^{-1}} = 10mA$

When only switch k_2 is closed then $I_{\max} = \frac{10\times10^{-3}}{10\times2\times10^{-1}} = 5mA$

When both the switches are closed then maximum current through L_2 is

$$I_2 = Q_0\sqrt{\frac{L_1}{L_2(L_1+L_2)C}} = \sqrt{5}mA$$

$\therefore$ **(b, c, d)**

11. Displacement of image of particle *A* is in phase with the displacement of particle *A* hence the displacement equation of particle *A*'s image is

$$Y_A' = 0.2\sin(\pi t) \qquad \ldots (1)$$

Displacement of image of particle *B* is out of phase with the displacement of particle *B* therefore the displacement of equation of particle *B*'s image is

$$Y_B' = 0.2\sin(\pi t + \pi) \qquad \ldots (2)$$

Relative vertical displacement of *A* w.r.t. to *B* is

$$Y_{AB}' = Y_A' - Y_B'$$

$$= 0.2\sin(\pi t) - 0.2\sin(\pi t + \pi)$$

$$= 0.4\sin(\pi t) \qquad \ldots (3)$$

$\therefore$ **(c, d)**

12. Fringe width $= \dfrac{\lambda D}{d}$

If D increases then fringe width increases and fringes inflate λ decreases when the whole system is immersed in a liquid of refractive index μ then fringe width decreases fringes increases.

$\therefore$ **(a, c, d)**

SECTION–III
Comprehension Type

13. $r = \dfrac{q_1 q_2}{4\pi\varepsilon_0 (2KE)}$

$v^2 = \dfrac{q_1 q_2}{4\pi\varepsilon_0 mr}$

$v^2 = \dfrac{1.6\times10^{-19}\times1.6\times10^{-19}\times9\times10^{9}}{10^{-15}\times2\times1.67\times10^{-27}}$

$v = 8.3 \times 10^6$ ms^{-1}

$\therefore$ **(d)**

14. $R = \dfrac{mv}{eB}$

or $B = \dfrac{2\times1.67\times10^{-27}\times8.3\times10^{6}}{1.6\times10^{-19}\times1.25} = 139mT$

$\therefore$ **(c)**

15. $\Delta E_1 + \Delta E_2 = \dfrac{(0.023213)\times1.66\times10^{-27}\times(3\times10^{8})^2}{1.6\times10^{-19}} = 21.6MeV$

$\therefore$ **(b)**

SECTION–IV
Matrix–Match Type

1. **(A) – (1, 2, 3, 4); (B) – (1, 3, 4); (C) – (3, 5); (D) – (1, 2, 3, 4)**

2. **(A) – (1, 2, 3, 5); (B) – (1, 2, 3, 5); (C) – (1, 4, 5); (D) – (1, 4, 5)**

SECTION–V
Integer Answer Type

1. Work done by friction is simply the changes in total energy of the system as the particle moves from point B to point C

$W_f = T.E._i - T.E_f = MgR\cos37^\circ - (MgR\cos60^\circ)$

$W_f = 0.3MgR$

$W_f = 6\,\text{J}$

Ans.: 6

2. Let the piston be displaced by x**.**

$$PA + \frac{q^2}{2\varepsilon_0 A} = P_0 A$$

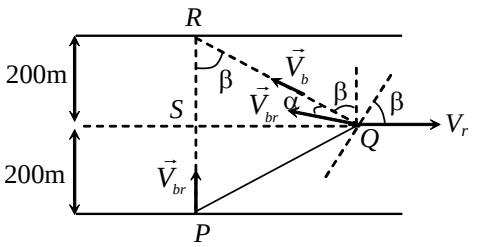
$$PA + \frac{1}{2\varepsilon_0 A}\frac{\varepsilon_0^2 A^2}{(L-x)^2}\varepsilon^2 = P_0 A$$

$$P = P_0 - \frac{\varepsilon_0 \varepsilon^2}{2(L-x)^2} = P_0 - \frac{2\varepsilon_0 \varepsilon^2}{L^2}$$

$\therefore \quad n = 2$

Ans.: 2

3. To reach point R from Q, the boat should be aimed at point R as seen from ground



$$\vec{V}_r \cos\beta = \vec{V}_{br} \sin\alpha$$

$$2 \times \frac{2}{\sqrt{5}} = 4\sin\alpha$$

$$\sin\alpha = \frac{1}{\sqrt{5}}$$

$$\Rightarrow \quad \tan\alpha = \frac{1}{2}$$

Also $\quad \tan\beta = \frac{1}{2}$

$$\theta = \alpha + \beta$$

$$\Rightarrow \quad \tan\theta = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{4}{3}$$

$\therefore \quad n = 3$

Ans.: 3

4. At steady state the rate at which the liquid is entering the tank should be equal to the rate at which the liquid is flowing out of the tank therefore

$$A\sqrt{2gh} = 2Av$$

$$h = \frac{2v^2}{g} = 5\,\text{m}$$

Ans.: 5